

Warm Up:

1. $\frac{1}{4} + \frac{3}{8}$
2. $4.5 - 5.2$
3. $-14.5 - (-12.8)$
4. $-2 + \frac{2}{3}$
5. On Friday morning, Hanna had \$235 in her checking account. On Friday afternoon, she used her debit card to spend \$42, \$10, and \$35. What is the balance in Hanna's account after these purchases?

1.2 Homework Questions?

Speak Like A Mathematician:

additive identity:

multiplicative identity:

multiplicative inverses:

reciprocals:

Key Concept — Properties of Equality

Reflexive Property	
Words:	
Symbols:	
Examples:	

Symmetric Property	
Words:	
Symbols:	
Examples:	

Transitive Property	
Words:	
Symbols:	
Examples:	

Example 1: Identify Properties of Equality

- If $13 + 25 = 38$, then $38 = 13 + 25$.
- $y + 4 = y + 4$
- $22 + 7 = 22 + 7$
- If $36 = 17 + 19$, then $17 + 19 = 36$.

Example 2: Interpret Properties of Equality

- If the amount of sugar in a recipe is equal to the amount of flour plus 2 tablespoons, and the amount of flour plus 2 tablespoons is equal to the amount of milk, then the amount of sugar is equal to the amount of milk. Let a = the amount of sugar, b = the amount of flour plus 2 tablespoons, and c = the amount of milk.

A) Write an algebraic statement for this scenario.

B) Which property of equality makes this statement true?

Example 3: Use Properties of Equality

- $y + 21 = \underline{\hspace{2cm}}$; Reflexive

- If $24 + 11 = 9 + 26$ and $9 + 26 = z$, then $24 + 11 = \underline{\hspace{2cm}}$;
Transitive

Key Concept — Addition and Multiplication Properties

Additive Identity	
Words:	
Symbols:	
Examples:	

Additive Inverse	
Words:	
Symbols:	
Examples:	

Multiplicative Identity**Words:****Symbols:****Examples:****Multiplicative Property of Zero****Words:****Symbols:****Examples:****Multiplicative Inverse****Words:****Symbols:****Examples:****Associative Property****Words:****Symbols:****Examples:****Commutative Property****Words:****Symbols:****Examples:**

Examples 4, 5, and 6: Evaluate Using Properties

- $4 - 2^2 + 8(2)$
- $(\frac{3}{2})(\frac{2}{3})(7)$
- $(4 - 3(2) + 7) \cdot 0$

Example 7: Determine Which Property Was Used

- $114 + 71 + 19 + 26$
- $= 114 + (71 + 19) + 26$
- $= 114 + 90 + 26$
- $= 114 + 26 + 90$
- $= 140 + 90$
- $= 230$

Examples 8 and 9: Evaluate Using Commutative and Associative

- $(\frac{5}{6})(9)(\frac{6}{5})$
- Josefina needs to evaluate $(\frac{7}{3}) \cdot 2 \cdot 3$. How can she use the commutative property to make her evaluation easier?
- Jade buys groceries once each week and wants to know what she spent over 4 weeks: $32 + 27 + 28 + 33$. Use one of the properties to evaluate.
- Use the commutative and associative properties to evaluate $(\frac{5}{9}) \cdot 9 \cdot (\frac{9}{5}) \cdot 4$.